ComNav Technology

User Guide

Post-processing Software ---Compass solution

Version: V 2.0.1

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About this User Guide

Thank you for choosing our Post-processing Software - Compass Solution. This User Guide provides introduction of software installation, project management, data processing and reporting. It can help you to get familiarized with its interface and start your project effectively.

DISCLAIMER

Please read this software disclaimer carefully before installing Compass Solution.

1) Compass Solution software is an intelligent post-processing software owned by ComNav Technology Ltd.
2) ComNav declares that software has been intensively tested, but we make no warranty that it is compatible with all software or hardware systems, or error-free. We are not responsible for any damages, liability or injuries caused by any failure of software performance.

BY INSTALLING Compass Solution SOFTWARE, YOU AGREE TO ALL TERMS LISTED ABOVE. IF YOU DO NOT ACCEPT THE TERMS, PLEASE UNINSTALL IT IMMEDIATELY.

Technical Assistance

If you have any question and can't find the answer that you need in this manual, please contact your local dealer from which you purchased the M300Pro. Alternatively, please request technical support using the ComNav website at: www.comnavtech.com or technical support email: support@comnavtech.com.

Your Comments

Your feedback about Compass solution software and this User Guide, or other ComNav products will help us to improve with future revisions. Please email your comments to: support@comnavtech.com.

1 Introduction

Compass Solution software is post-processing software designed to process and analyze raw GNSS (Global Navigation Satellite System) data, obtaining high accuracy positioning solution. It can support most of the GNSS satellite constellations (GPS, GLONASS, Galileo and BeiDou/Compass), and main functions are shown below:

1) Process GNSS observation data in RINEX and ComNav formats.
2) Supports differential post-processing in static and kinematic modes.
3) Network adjustment of all processed baseline vectors.
4) Output analysis reports in various formats (HTML, DXF, TXT, and KML).

1.1 Installation

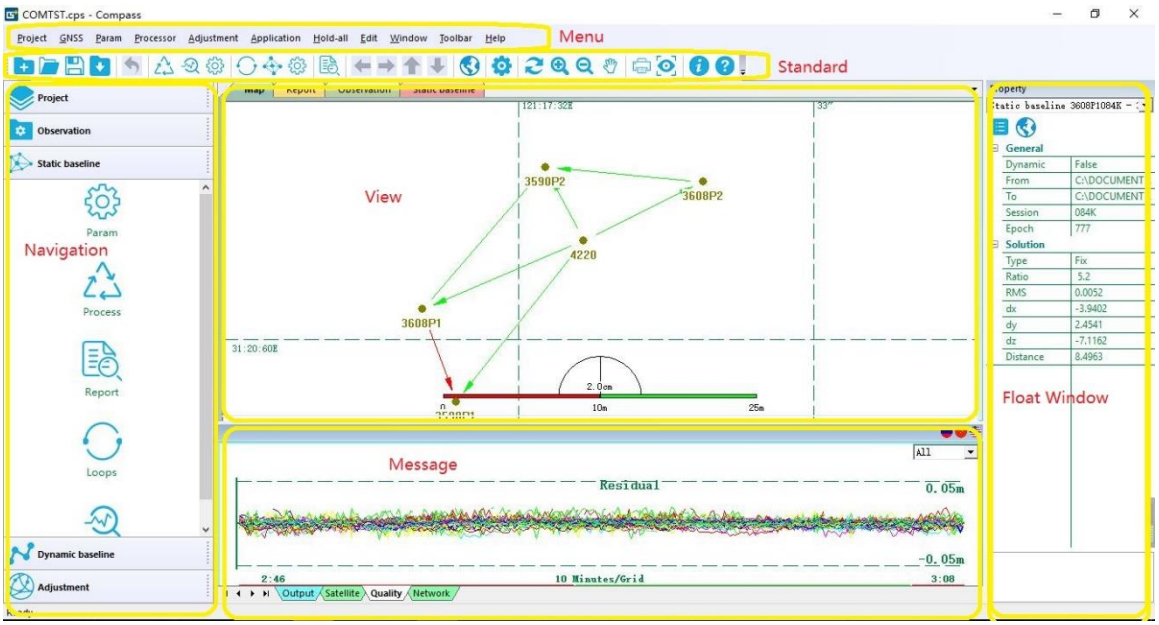
Double-click on compass setup.exe, choose install path to install the software. Please make sure that the main program, dongle and C++2012 are installed completely.

The operation system should be at least or better than:
Windows 2000
CPU: core 2.0GHz;
Internal memory > 1G, HD>80 G;
Screen resolution: 1280*1024.

1.2 User Interface

This section describes the user interface and main menus of Compass Solution software.

- Menu Bar: contains all functional menus.
- Standard Toolbar: quick access the frequently used functions, including import observation files, baseline processing, network adjustment
- View Area: the main display window.
- Navigation Pane: navigate to different views of your raw and processed GNSS data.
- Float window: shows the information of raw or processed data, go to menu Bar -> View -> property to active float window.
- Message box: contains the output message, timelines and quality of observation.



1.2.1 Menu Bar

The menu bar includes File, Edit, GNSS, Processor, Adjustment, Report, Tools, View, Window and Help. The main functions of each menu are described below:



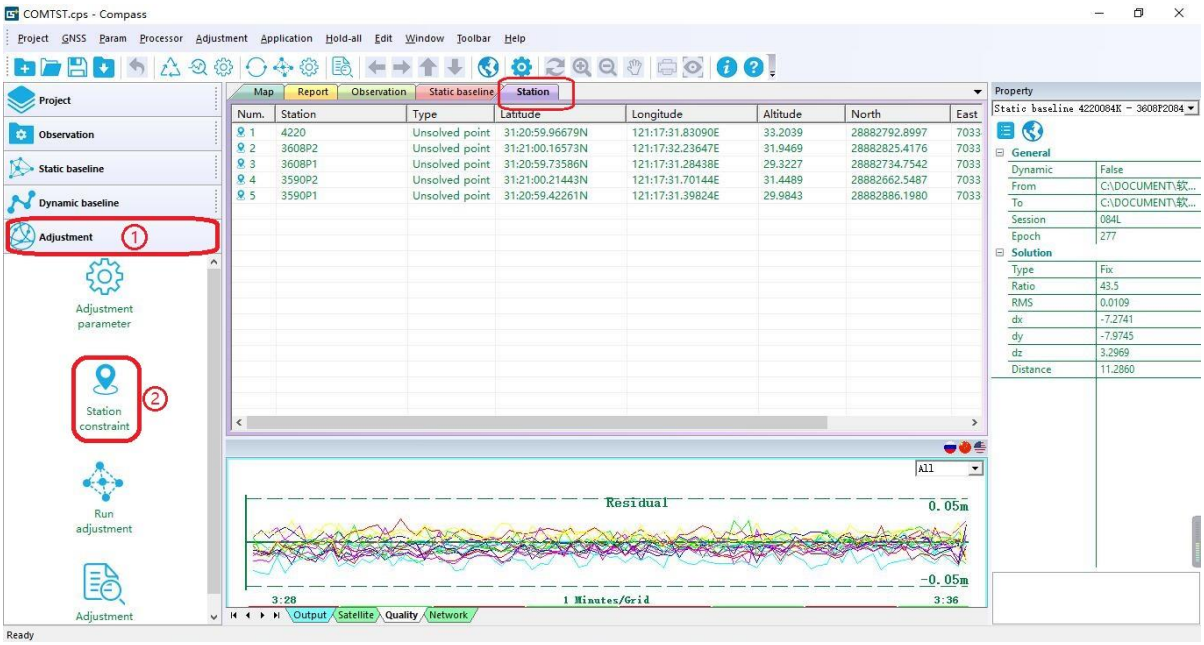
- Project: create, open, save and close the project, and import raw data.
- GNSS: GPS, Compass(Beidou) and GLONASS.
- Param: datum, antenna etc. unit configuration and processing configuration.
- Processor: process all or selected baselines, baseline settings.
- Adjustment: check observation files, check Network, search Errors, adjustment setup and run adjustment.
- Application: output analysis report in various formats, including DXF, TXT, KML.
- Hold-all: the 7-parameters, coordinate convert, file cutting and source path.
- Edit: undo, delete the selected items, and replace path and property of observation sites.
- Window: switch map, observation and report windows.
- Toolbar: customize and edit toolbar.
- Help: Help and about compass solution.

1.2.2 Navigation Pane

It contains five tabs: compass list, observation, dynamic baseline, static baseline and station.

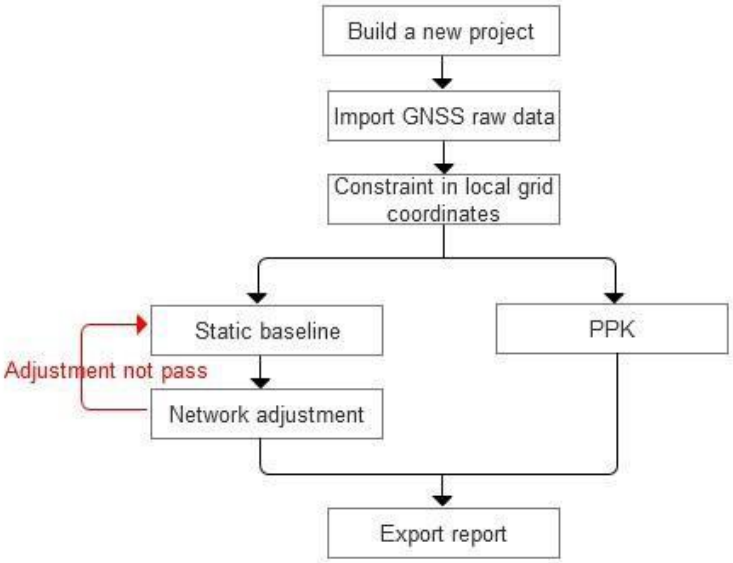
- Compass list: Map view of baseline (static baseline), and route of PPK will be shown after dynamic baseline processing.
- Observation: shows the detailed information of observation files.
- Dynamic baseline: dynamic baseline list
- Static baseline: static baseline list
- Station: the station marker list

To change the display in the main view area, you should firstly click ① navigation Pane, then select the particular display items in ② area, the selected items will be shown in the View area.



1.3 Workflow

The diagram shows the workflow of Compass solution software, which gives you a clear view of baseline processing.



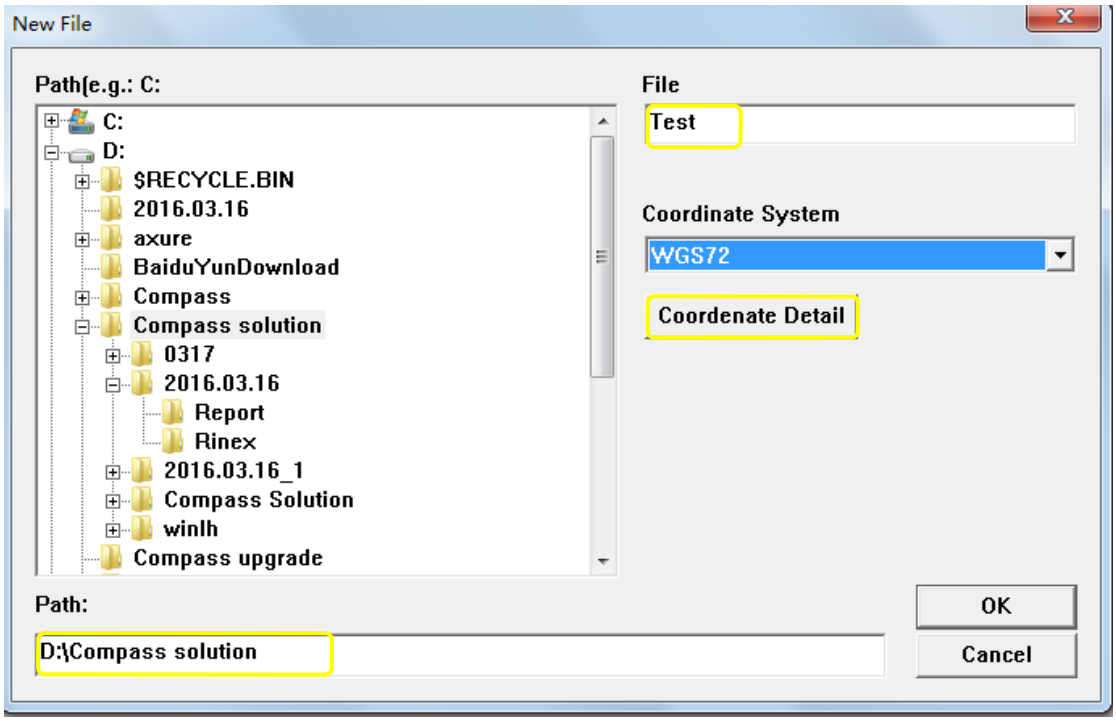
2 Project Management

In this chapter you will learn to create a new project, import raw data files and add coordinate constraints.

2.1 Create a new project

Firstly, you should install Compass solution software properly, and download GNSS raw data from the receiver to your PC. Then, run **Compass Solution** -> **click File** -> **New Project**, then:

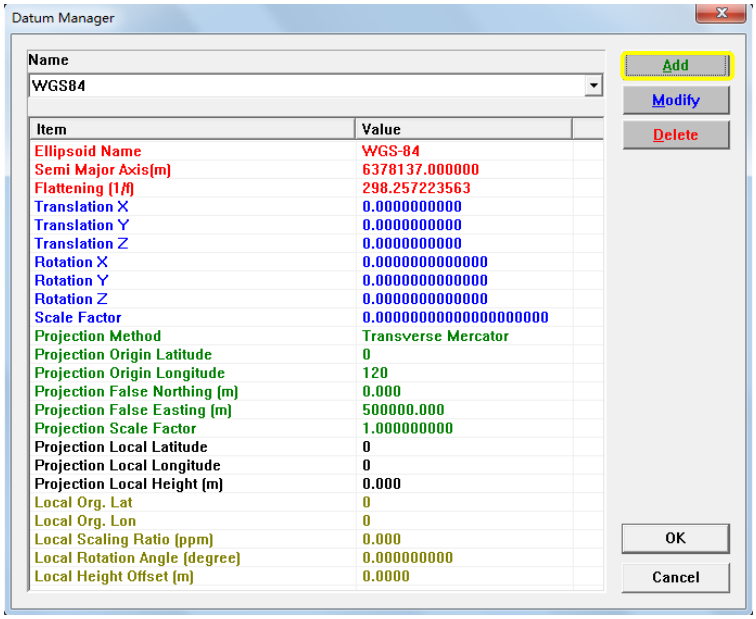
1) Choose the folder path
2) Enter a file name
3) Select coordinate System, you can select one from the coordinate system list (WGS72, WGS84, Xian80, Beijing54 and CGCS2000), or click coordinate detail to add a local datum system.



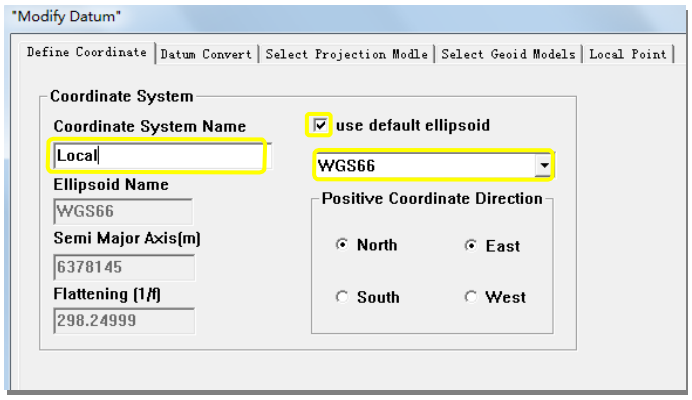
Tips: 1) For save path, we recommended that you save the project and GNSS raw data in the same path, which is easier for project management; 2) To download GNSS raw data to your PC, please reference to ***Compass receiver utility User Guide****.*

Click **Add** to define new datum, you can set proper parameters (ellipsoid and

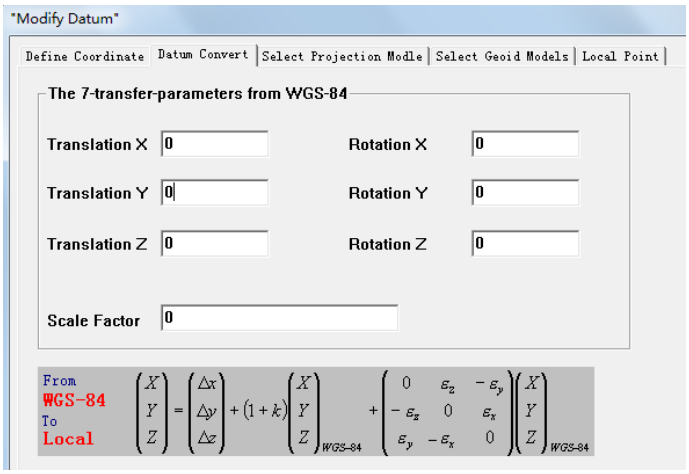
projection) for your new coordinate system; Also Click **Modify** to edit the currently coordinate parameters.



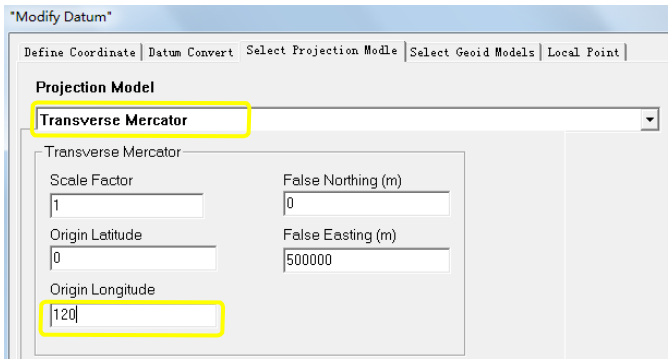
Firstly check "use default ellipsoid", select the proper ellipsoid from the list, and then enter the Coordinate System Name.



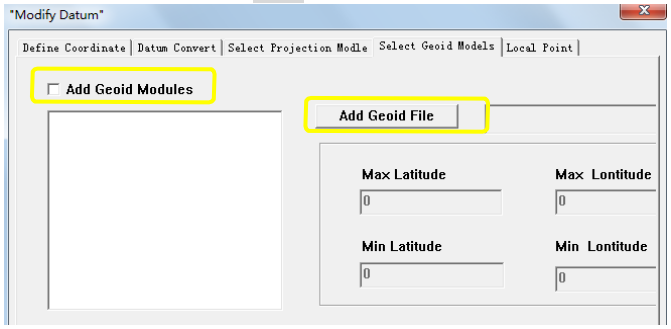
Enter 3 or 7 parameters if needed



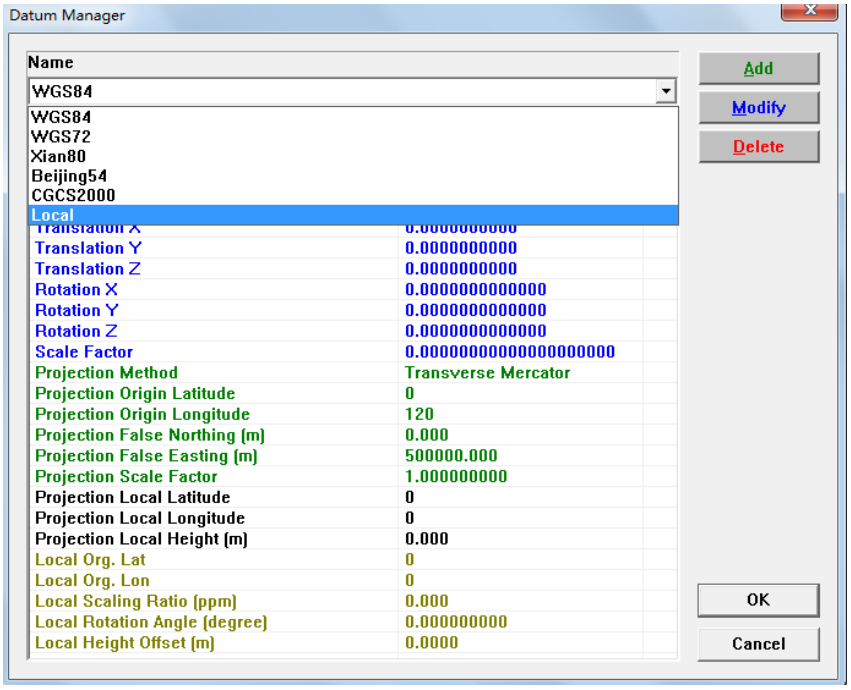
Select the proper projection model, and enter parameters according to the Project location.



In Geoid Models, check ***Add Geoid Modules***, click ***Add Geoid File*** to input a Geoid module. Then click ***OK*** to finish datum modify.



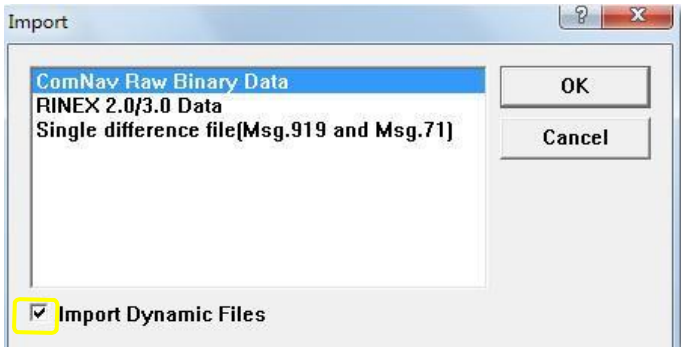
Then you can select the coordinate system you built, click ***OK*** to active this datum in the current project.



Tip: the coordinate system and its parameters also can be edited in ***Tools*** *->* ***datum manager****.*

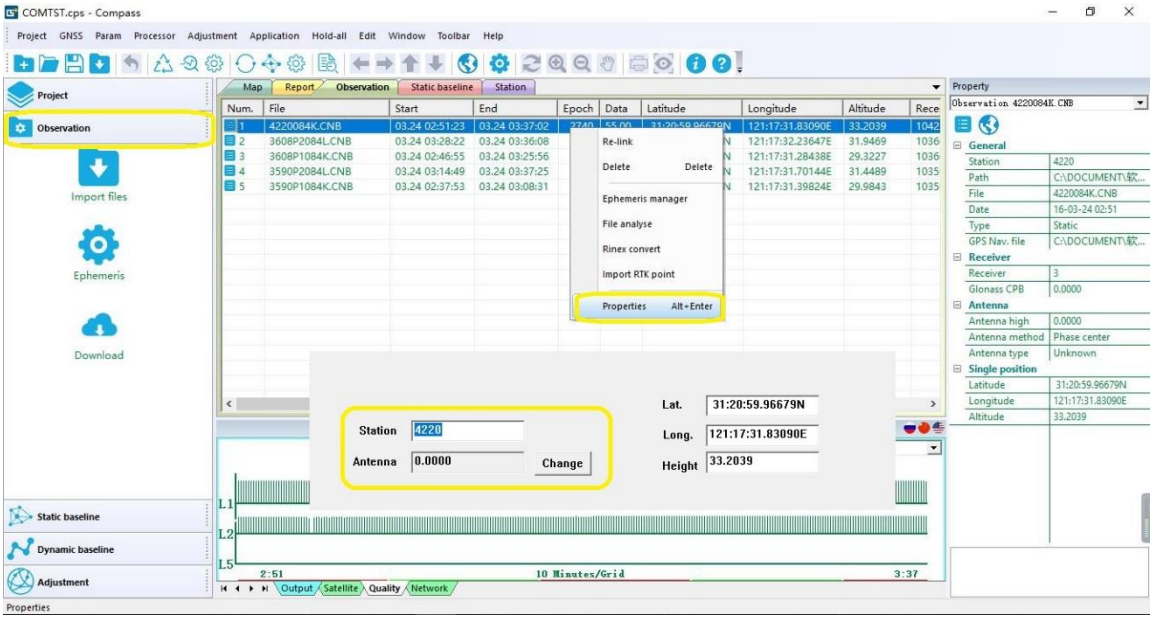
2.2 Import raw observation files

To import GNSS raw data, click ***File*** -> ***Import*** in the menu bar. The static observation file can be imported directly; For PPK and dynamic data, please check “Import Dynamic files”.



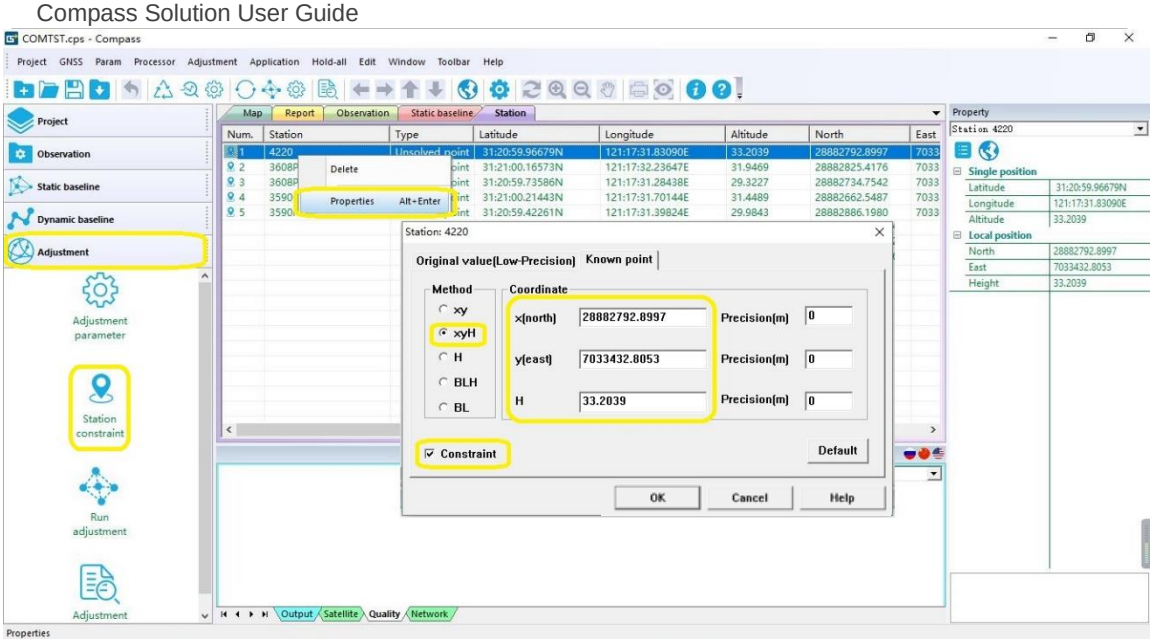
Tip: Compass solution only can support dynamic raw data in ComNav format.

After importing the raw data, click ***Observation*** -> ***select the observation file*** in the navigation pane, then right-click on the file name to edit properties, including station name and antenna information.



2.3 Constraint known points

To ensure all stations within your local grid coordinate system, at least 1 station should be fixed on a known point. Click ***station*** -> ***properties*** -> ***Known Point*** to enter coordinate values (check the ***Constraint***).



*Tip: You can get the default coordinate value under current coordinate system by clicking Default. It should not be a big difference between the **default** value and the **known** value; otherwise, check the coordinate system parameters.*

3 Baseline processing

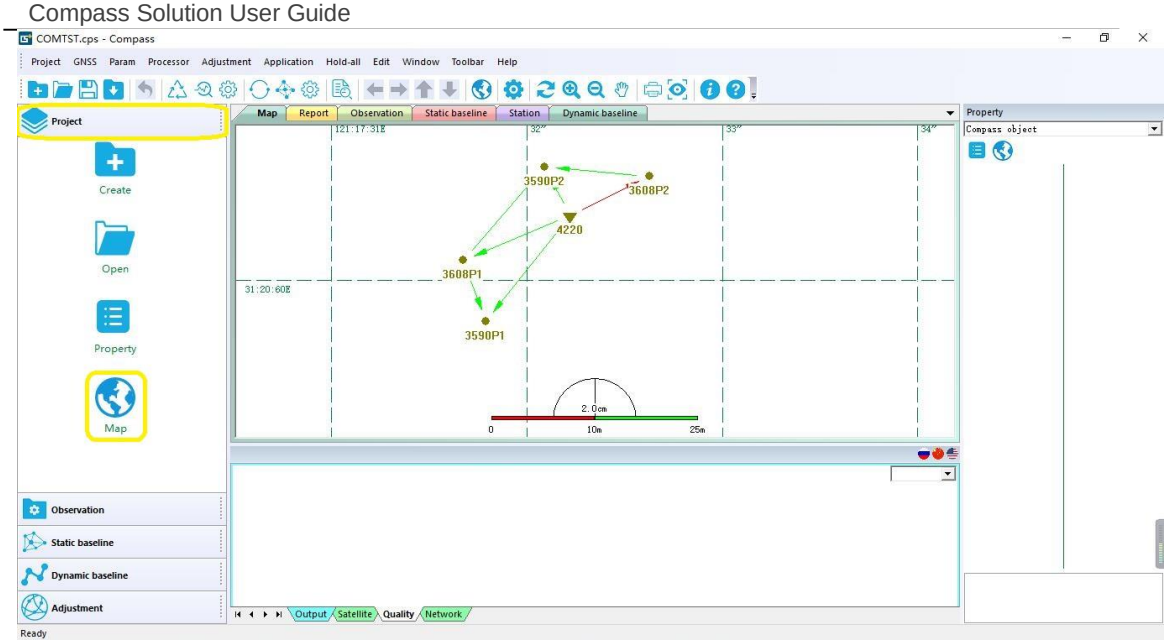
This chapter will introduce how to process GNSS raw data in static and PPK mode respectively, which helps you to learn about the basic settings in the software.

3.1 Static baseline processing

In static mode, Compass Solution can directly process baseline vectors of different sites with simultaneously recorded GNSS raw data. We recommended that at least 3 receivers observe at the same time to form the simultaneously loop of baselines.

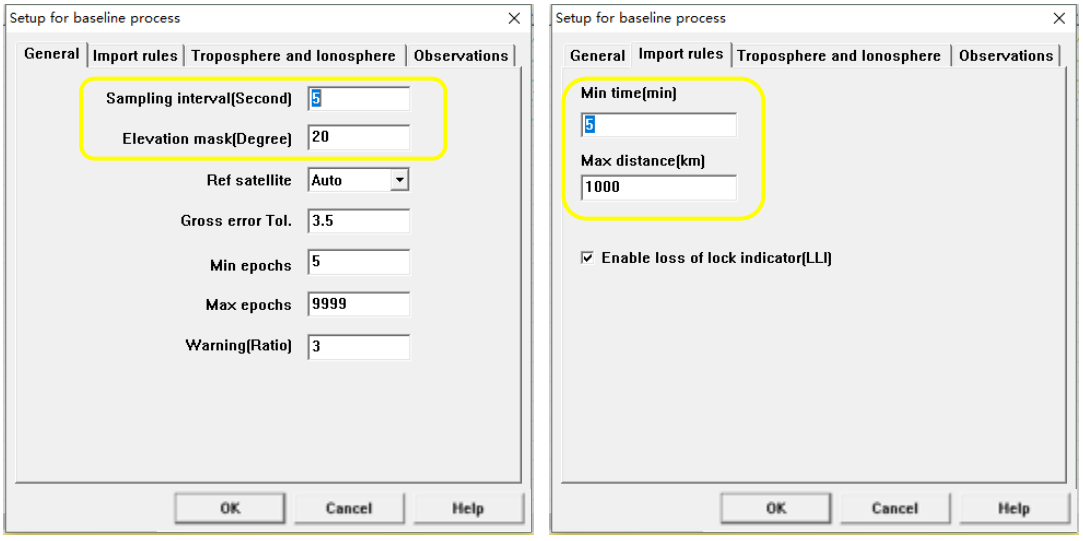
3.1.1 View

The static baseline vector will be automatically formed after importing GNSS raw data. Go to *Project* -> *Map* to check the baseline vectors.

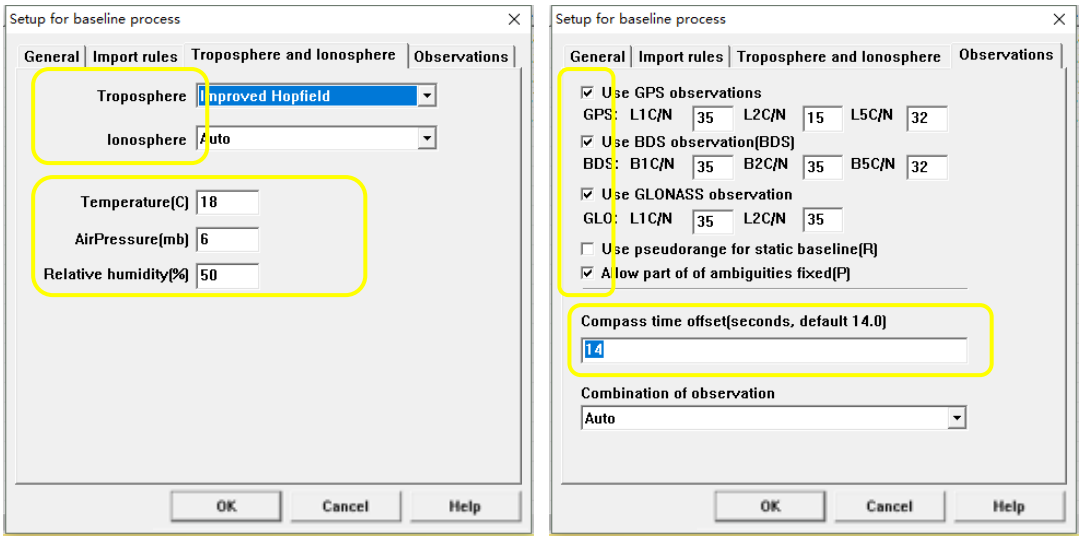


3.1.2 Baseline Process Settings

Before starting static baseline processing, click ***Processor*** -> ***Baseline setting*** in the menu bar to set proper processing parameters. The settings are related to data import rules, observations, troposphere and ionosphere.



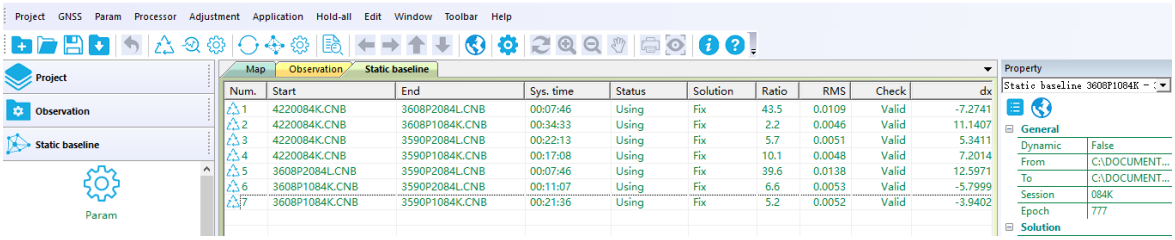
- **Sampling Interval:** the data processing interval, the value should be greater than the interval used during recording of the data.
- **Elevation Mask:** the minimum mask angle to be used. When the data is a good quality the mask should be between 10 and 15 degrees.
- **Min Time (Min.)**: the minimum observation period of GNSS raw data to form the baseline.
- **Max Distance (km)**: the maximum observation distance of GNSS raw data to form the baseline.



- Select proper correction model for Troposphere and Ionosphere.
- Weather parameters: Temperature, Airpressure and Relative.
- Allow you to disable satellites individually so that data from these satellites will not be part of the solution.
- Compass time offset: default value is 14.

3.1.3 Static Baseline process

After setting the baseline process parameters, *click* ***Processor*** -> ***Process Static Baselines***.



After all baseline processed, check the quality value (Solution type, Radio and RMS) and baseline information (dx,dy,dz,ds) shown in the view area. To acquire a high-accuracy result, also check:

- Solution value from Unsolved to Fix
- Radio should be smaller than warning value you set (default as 3)
- RMS value< 0.00X (millimeter accuracy)
- Check: valid

Alternatively, click ***Adjustment*** -> ***Search Loop's Errors*** to check PPM value of loop and detailed baseline processing information.

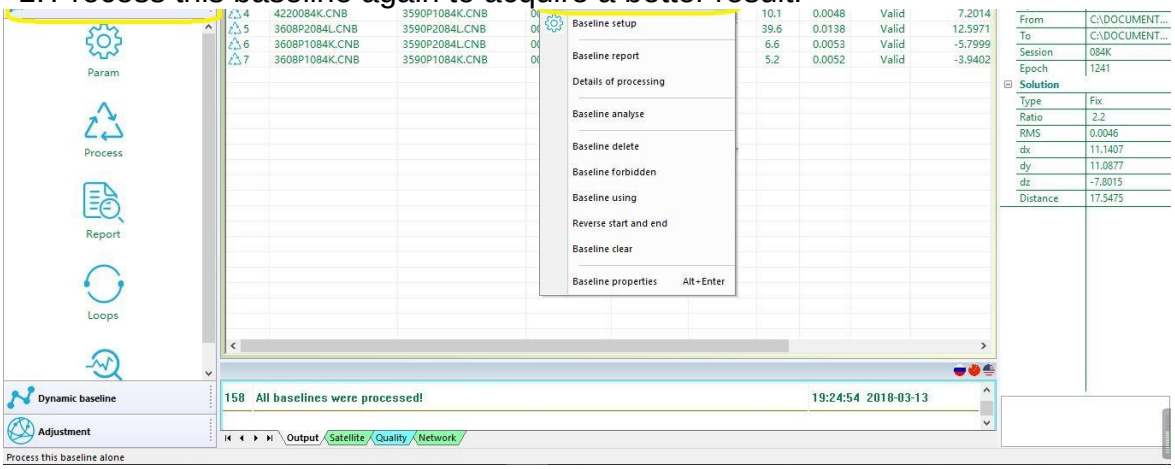
If the baseline RMS and Ratio values cannot meet standard requirements, edit the baseline manually and process it again.

Delete bad observation data in the *Timeline window* based on the *Quality window*. As the below example shown, observation data with a bad quality should be deleted from Timeline window.



Tip: In the Time Line window, selecting the observation means delete the observation data, double choose left checkbox to clear the selecting to restore the deleted observations.

1. Process this baseline again to acquire a better result.



You can also edit baselines, including:

- Baseline setup: shortcut to baseline processing setting.
- Baseline report: create report for baseline processing result.
- Details of processing: show details of calculation procedure parameter.
- Baseline delete: delete this baseline from your project, which is not recommended.
- Baseline forbidden: forbid this baseline to be part of the solution, used only when the baseline without good accuracy.
- Baseline using: allow this baseline to be part of the network adjustment
- Reverse start and end: reverse the direction of baseline vector
- Baseline properties: include general information, fix solution and float solution.

3.1.4 Network Adjustment

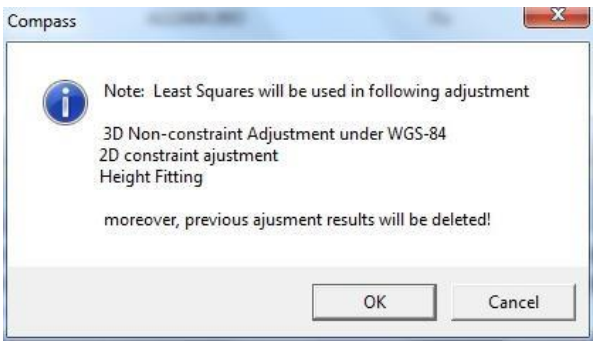
In this section, you will carry out a least squares adjustment using the processed baselines above. Firstly go to ***Adjustment*** -> ***Adjustment setup*** in the menu bar.



- 3D: WGS84 coordinate as control points
- 2D: Horizontal adjustment, use local gird coordinates as control points
- Height Fitting: Height adjustment

Tip: we recommend that you apply 2D & Height to adjust baselines.

Then click ***Adjustment***->***Run Adjustment***, click ***OK*** to complete adjustment.

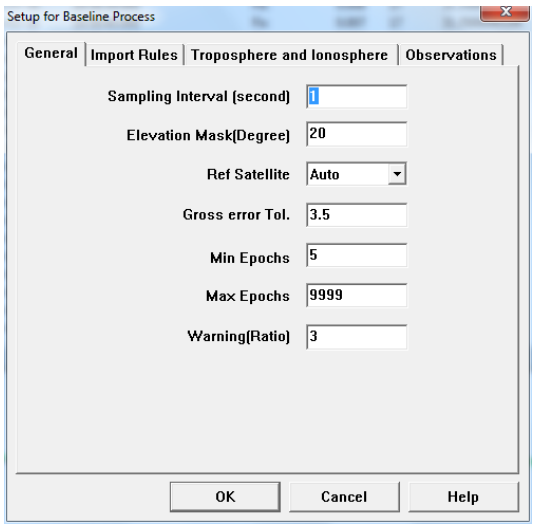


3.2 PPK (Post-Processing Kinematic)

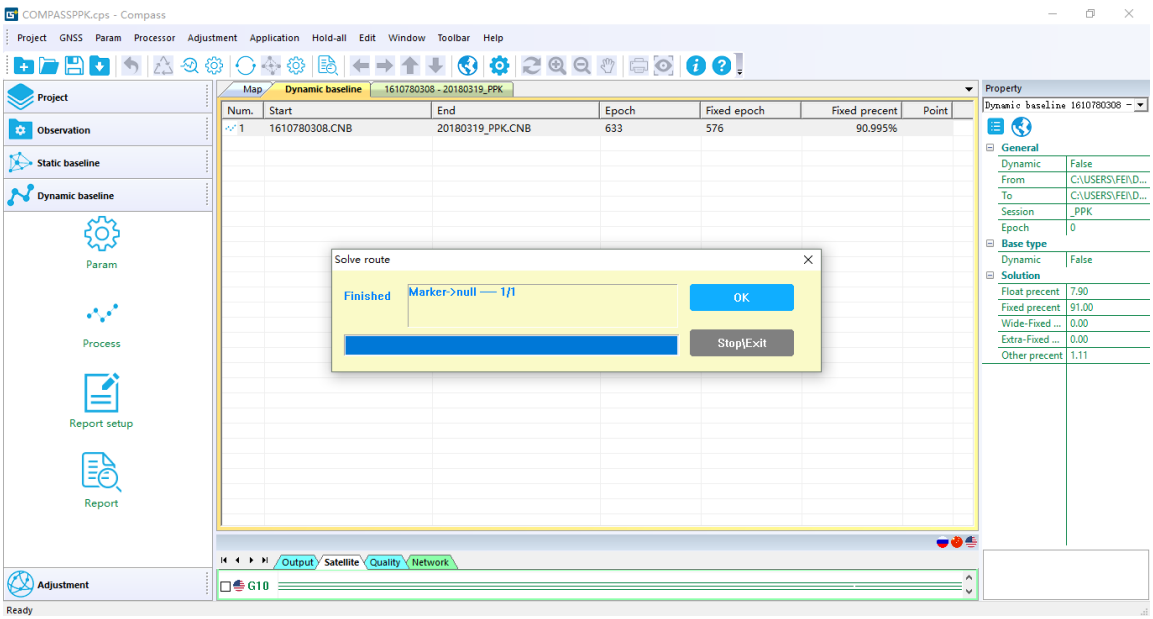
For PPK (Post processed Kinematic) mode, it can process trajectories of rover combined with base station static data. Commonly, set the sampling interval of base station as 1/s and fix it on the known point *(see 2.3 Constraint known points)*.The dongle is needed to active the PPK function, make sure the dongle is connecting to your PC and installed the drive properly.

3.2.1 Dynamic baseline processing

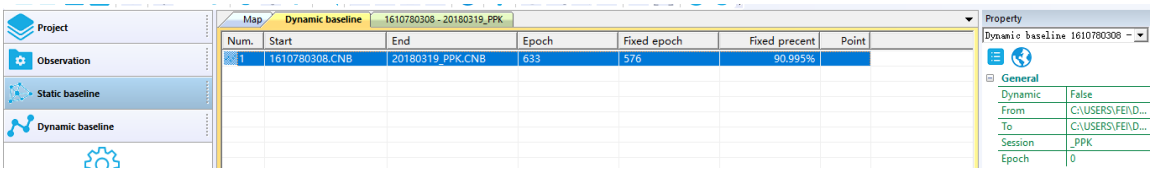
Setting up process parameters for the dynamics baseline, which should be the same as static baseline processing setting (the sampling interval is 1/s).



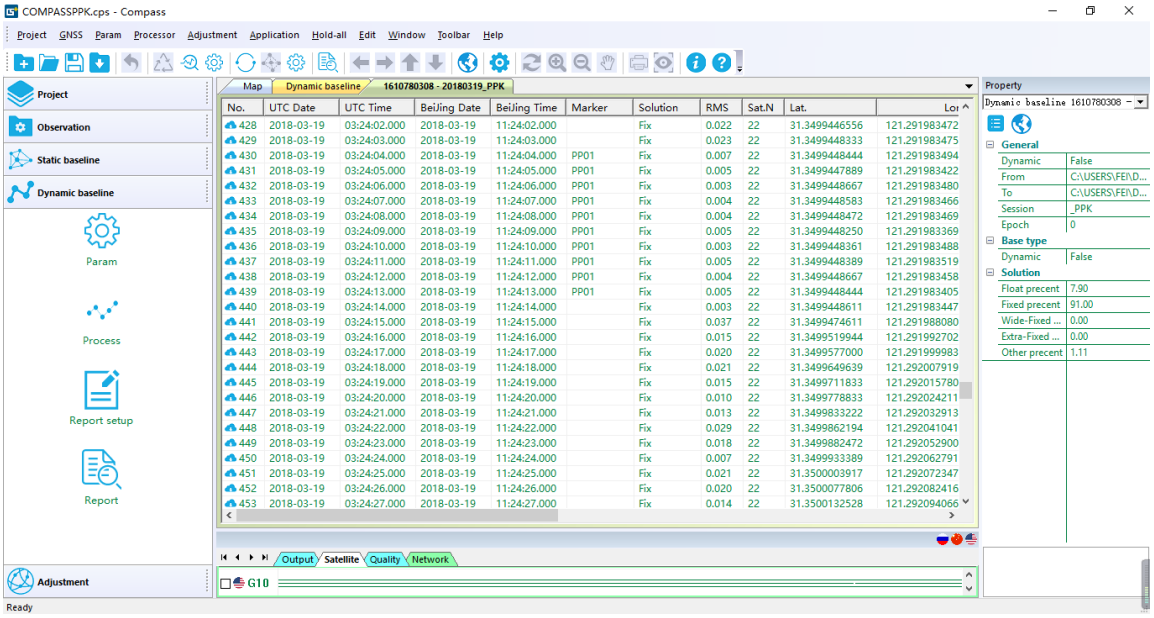
Right-click on the file's name to process this baseline alone. The figure is shown below:



Click ***Next*** to finish dynamic baseline processing, you will see the baseline information, i.e. Epoch Num, Fixed Epoch and Fixed percentage.

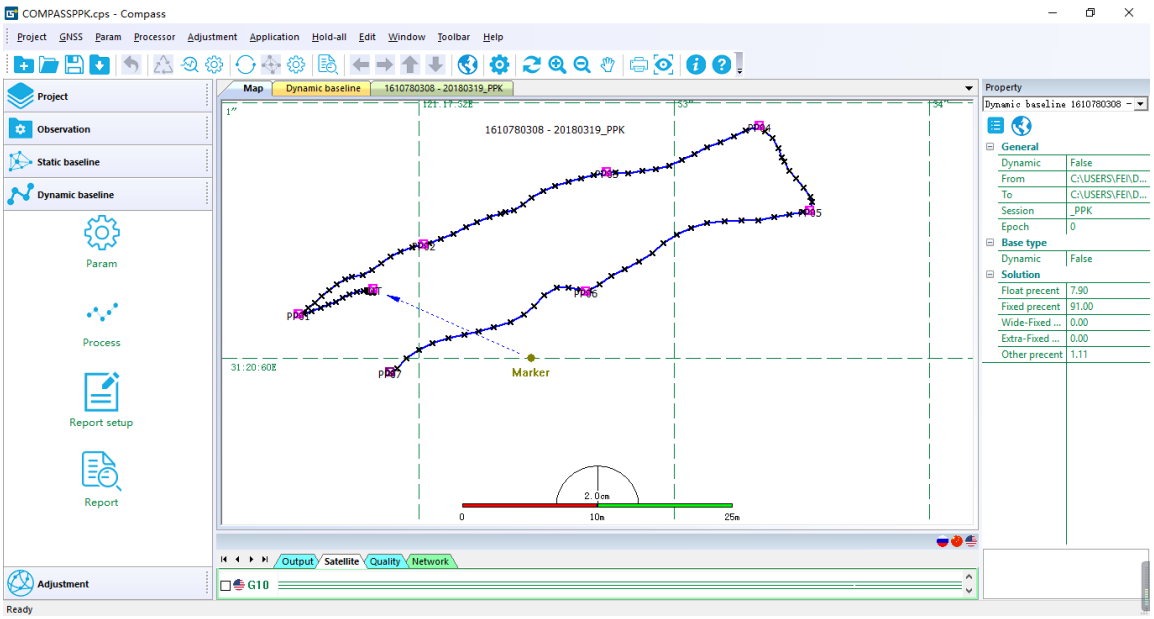


Double right click on the Dynamic baseline to check detailed information of each point.



3.2.2 Map View

After dynamic baseline processing, you can go to ***Project*** -> ***Map*** to have a look at trajectories of your rover.



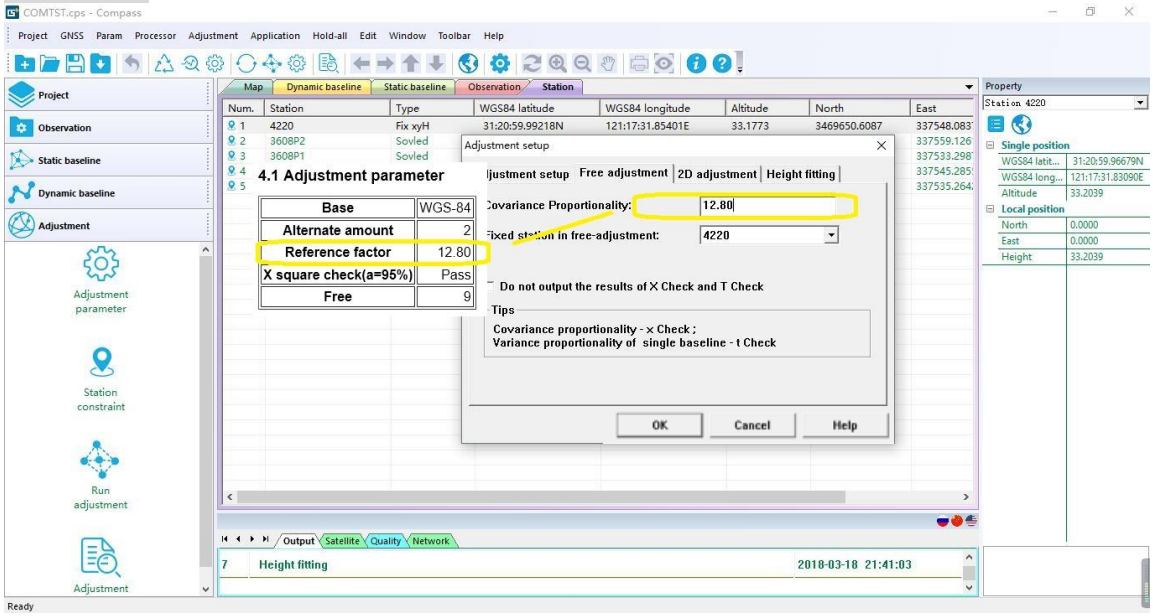
4 Export the report

This chapter will introduce how to export processing reports.

4.1 Static Report

Go to ***Report*** in the main menu bar, you can output a network adjustment report in web format. It contains five parts: project property, coordinate system, baseline, 3D non-constraint adjustment and 3D constraint adjustment, which help you to check the result of baselines and adjustments.

The most important parameter you should check is ***X square check***, if it shows fail, copy the value of reference factor to ***Adjustment*** -> ***Adjustment Setup*** -> ***Free adjustment***, then run adjustment again.

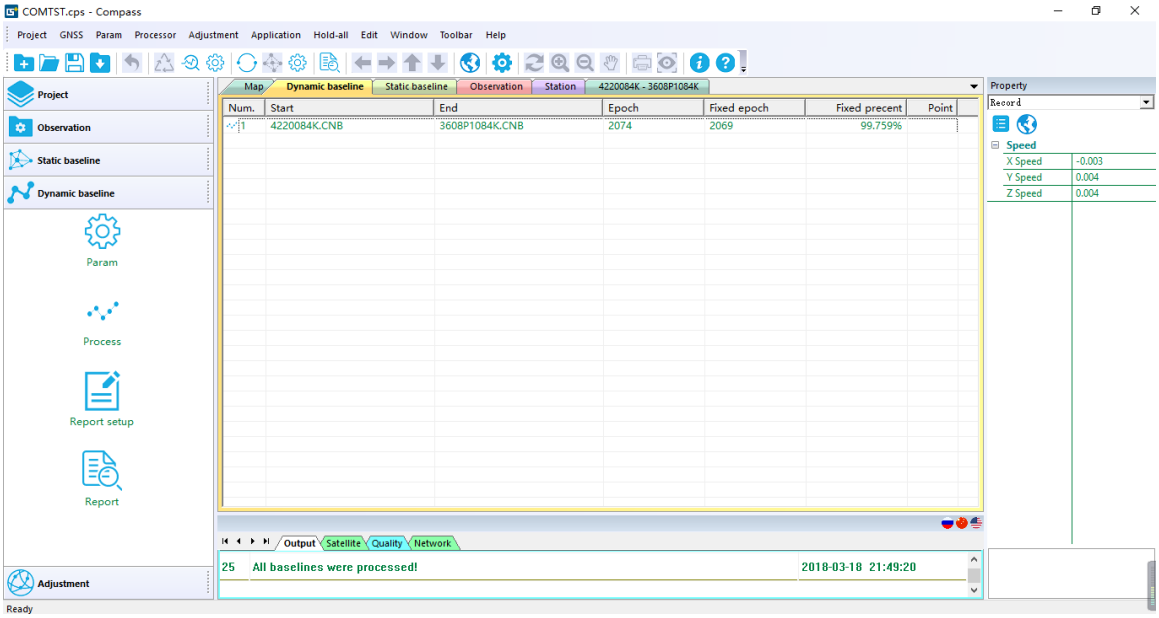


*Tip: the **X square check** parameter is meaningful for triangle network adjustment. For single baseline processing, you only need to check parameters, i.e. Solution, Ratio and RMS.*

Other report formats can be exported, including brief report, DXF and KML. In the end, click ***File*** -> ***Save*** to save your project.

4.2 PPK Report

Double-click on this baseline to check the detailed information.

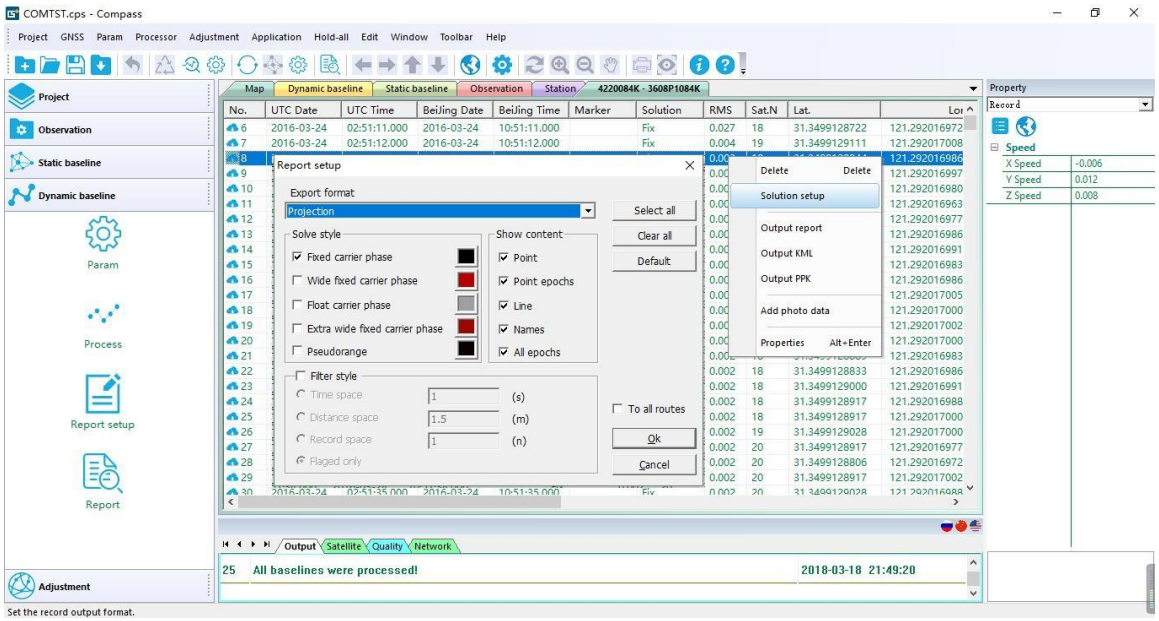


In the detailed information page, right click to select *Filter* -> *Export format,* *Solution Setup* -> *OK* to set the report format.

Export format:

- Latitude/longitude: export the report in WGS84 coordinate system
- Projection: export the report in local grid coordinate system.
- Relative 2D: Horizontal distance relative to base station
- Relative distance: 3D distance relative to base station

Solve style: Filter the processing results, commonly choose Fixed carrier phase (centimeter accuracy).



a) Output report, PPK report or KML format, we recommend to save the report in **.csv format, which makes it easier to analyze post-processing result. And the KML allows you to save files in the format accepted by Google Earth

b) After exporting the report, go to the main menu bar -> ***File*** -> ***Save***, save the project.